CLASS 8 — MATHS · FORMULA SHEET

Chapter 1: **A Square and a Cube** | Quick revision, one page

Squares and square roots

- A perfect square is n^2 for a natural number n; its units digit is 0, 1, 4, 5, 6 or 9 only.
- A number is a perfect square if every prime factor occurs an even number of times.
- $1 + 3 + 5 + \cdots + (2n - 1) = n^2$ (sum of the first n odd numbers).
- $(n + 1)^2 - n^2 = 2n + 1$; there are $2n$ non-square numbers between n^2 and $(n + 1)^2$.
- Square root by prime factorisation: pair equal primes and take one from each pair; $\sqrt{a \times b} = \sqrt{a} \times \sqrt{b}$.

Patterns with squares

- $(2n)^2$ is even and $(2n + 1)^2$ is odd; a perfect square ends in an even number of zeros.
- $\sqrt{\dfrac{a}{b}} = \dfrac{\sqrt{a}}{\sqrt{b}}$; a square of a number ending in 5 ends in 25.

Cubes and cube roots

- A perfect cube is n^3; every prime factor occurs a multiple of 3 times.
- Cube root by prime factorisation: group equal primes in triples, take one from each triple.
- $\sqrt[3]{-a} = -\sqrt[3]{a}$; the cube of a negative number is negative; $\sqrt[3]{ab} = \sqrt[3]{a}\,\sqrt[3]{b}$.
- Cube of side a: volume a^3, surface area $6a^2$.

Special numbers

- $1729 = 1^3 + 12^3 = 9^3 + 10^3$ is the smallest number expressible as the sum of two cubes in two ways (Ramanujan's taxicab number).
- To make n a perfect square (or cube): multiply or divide by the primes that have an odd (or non-multiple-of-3) exponent.

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